

WHERE ARE WE GOING?

UNIT 1

FOUNDATIONAL CHEMISTRY SKILLS

Chapters 1 – 5

Weeks 1 – 4

Subatomic particles
Atoms, molecules, ion
Periodic table of elements
Avogadro's and the mole
Chemical formulas
Chemical equations
Stoichiometry

UNIT 2

HOW REACTIONS HAPPEN

Chapters 7, 14 – 16

Weeks 5 – 7

Gases
Kinetic molecular theory
Kinetics
Reaction mechanisms
Equilibria

UNIT 3

WHAT REACTIONS PROVIDE

Chapters 6, 18, 19

Weeks 8 – 10

Heat and enthalpy
Work
Gibbs free energy
Electrochemistry

UNIT 4

MICROSCOPIC TO MACROSCOPIC

Chapters 8 – 13

Weeks 12 – 16

Quantum model of atom
Orbitals
Electron configurations
Periodic trends
Lewis structures + VSEPR
Bonding theory
Molecular polarity
Intermolecular forces
Properties and solubility

06A THERMOCHEMISTRY

UNIT 03: WHAT REACTIONS PROVIDE

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 6

1. Define and apply energy, work, and heat.
2. Describe and apply the concept of conservation of energy.
3. Define and apply state functions and path functions.
4. Apply the concept of state functions to predict the energy changes involved in reactions.
5. Summarize the roles of enthalpy and heat flow in chemical reactions.
6. Calculate pressure-volume work.
7. Calculate the heat required to change the temperature of a substance.
8. Calculate changes in enthalpy and internal energy in chemical reactions.
9. Apply Hess's law to calculate the enthalpy of different reactions.
10. Apply standard enthalpies of formation and Hess's law to calculate the enthalpy of a reaction.

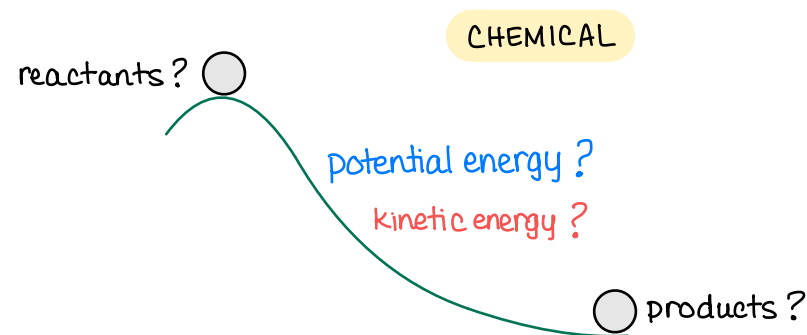
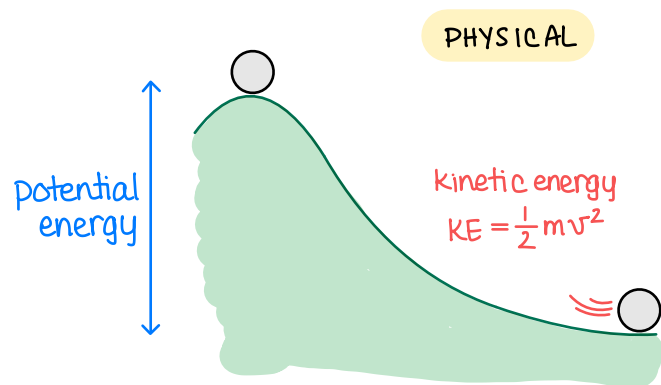
Energy and Energy Units

6.1

Thermochemistry: The study of the *energetic* consequences of chemistry.

We are often more familiar with *mechanical* energy, which is the energy due to the position and/or motion of an object.

- 1) Kinetic energy is energy of motion.
- 2) Potential energy is energy related to position.



Energy and Energy Units

6.1

More broadly, though, energy can be defined as

- The capacity to do work or
- The capacity to transfer heat

The SI unit of energy is the Joule (J):

- $1 \text{ J} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$
- $1 \text{ L} \cdot \text{atm} = 101.325 \text{ J}$
- $1 \text{ calorie} = 4.184 \text{ J}$

Energy, Heat, and Work

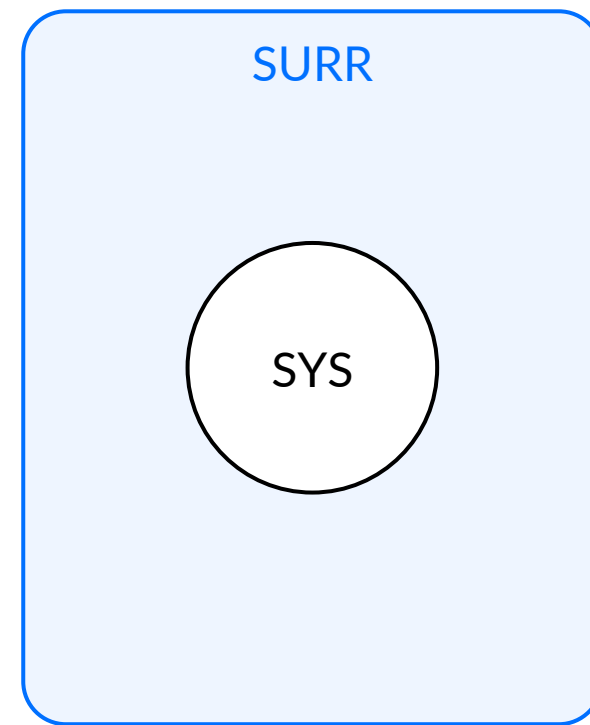
6.2

To facilitate our study of energy, we get/need to *define* what we are studying.

System = Region of interest often, a chemical reaction

Surrounding = Everything else that is *not the system*

Universe = Sys + surr



Energy, Heat, and Work

6.2

There are different kinds of *systems*.

- Open system: Both matter and energy can move between the system and the surroundings.
- Closed system: Energy but not matter can move between the system and the surroundings.

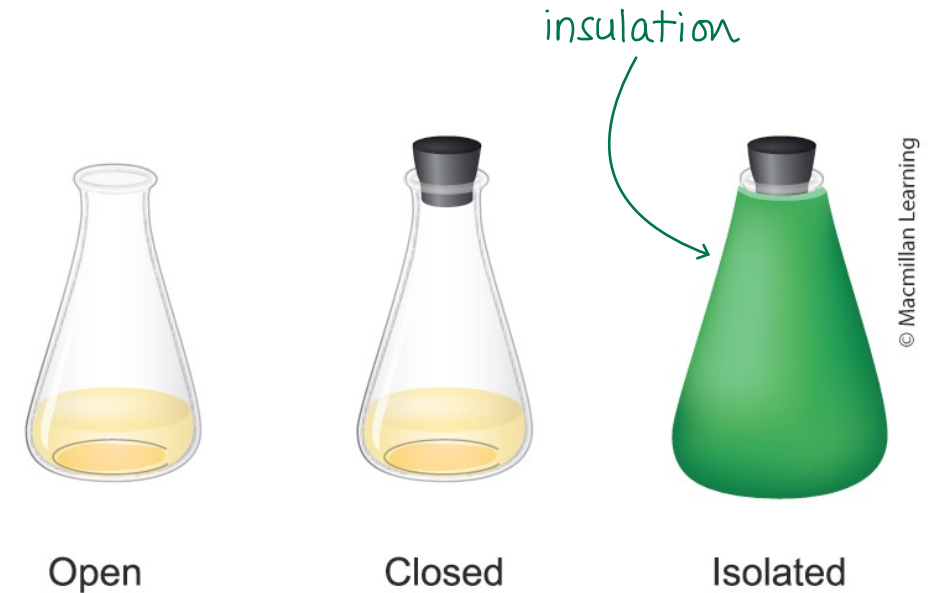


FIGURE 6.3

- Isolated system*: Neither matter nor energy can leave or enter the system.

*Note: The **universe** is the quintessential example of an isolated system.

Energy, Heat, and Work

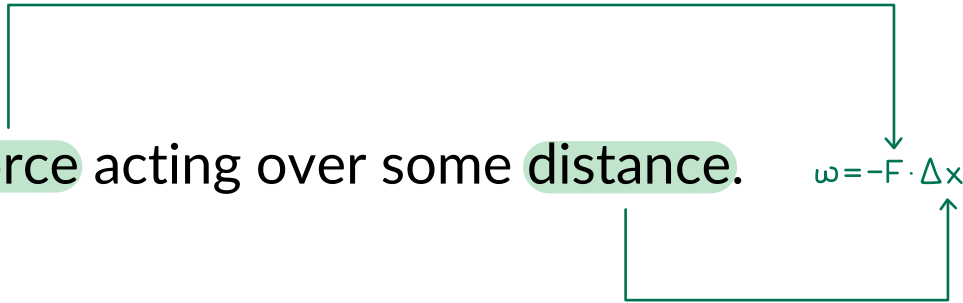
6.2

Energy is defined as the capacity to *do* work or the capacity to *transfer* heat.

- Note that both work and heat are *processes* where energy is *transferred*.

Work (w)

- Transfer of energy due to a **force** acting over some **distance**.


$$w = -F \cdot \Delta x$$

Heat (q)

- Transfer of energy due to a **difference in temperature** between objects.
- *Misconception:* Temperature is not the same as heat.

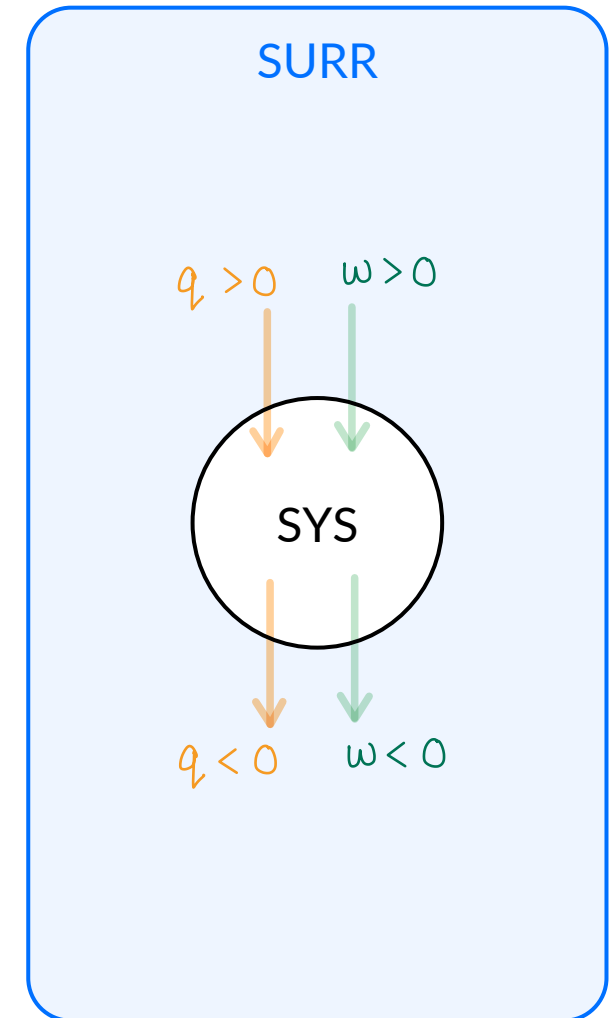
Energy, Heat, and Work

6.2

Because **work** (w) and **heat** (q) are both *transfers of energy* between the system and surroundings, we need to establish a consistent **sign convention** to denote energy changes.

→ **CAREFUL**: sign conventions change by discipline

Process	Energy of System	Sign
Heat <i>added</i> to the system	System <i>gains</i> energy	$q > 0$ $\oplus$
Heat <i>released</i> by the system	System <i>loses</i> energy	$q < 0$ $\ominus$
Work <i>done on</i> the system	System <i>gains</i> energy	$w > 0$ $\oplus$
Work <i>done by</i> the system	System <i>loses</i> energy	$w < 0$ $\ominus$



Energy, Heat, and Work

6.2

First law of thermodynamics: (in its different forms)

- For an isolated system (e.g. the universe), the total energy is *constant*.
- Energy cannot be created or destroyed, but it can be transformed from one form to another form.
- For a closed system, any changes in **internal energy (ΔU)** arise from *transfers* of energy with the surroundings via *only* **heat (q)** and/or **work (w)**.

$$\Delta U_{\text{sys}} = q + w$$

Internal energy (U and ΔU)

- **U** : The sum of the kinetic and potential energies of *all* particles in a system.
- **ΔU** : Only *changes* in internal energy can be measured.

IN-CLASS QUESTION 1

Learning Objective(s): 6.2

For a particular process $q = -10 \text{ kJ}$ and $w = +25 \text{ kJ}$.

sys releases heat to surr

work done on sys by surr

Which statement is **true**?

opposite

☐ Heat flows from the surroundings to the system.

☐ The system ~~does~~ work on the surroundings.

☐ $\Delta U = -35 \text{ kJ}$

$$\begin{aligned}\Delta U_{\text{sys}} &= q + w \\ &= -10 \text{ kJ} + 25 \text{ kJ}\end{aligned}$$

$$\Delta U_{\text{sys}} = +15 \text{ kJ}$$

☐ All these statements are true.

☒ None of these statements is true.

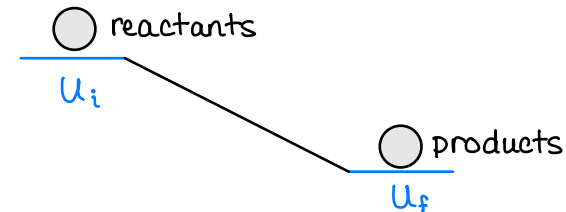
Energy as a State Function

6.3

State function

- Function whose value depends *only* on the (properties that define the) state of the system.
- More importantly, changes are *only* defined by the initial and final states and not the pathway between the two states.
- For example, internal energy (U) is a state function, so

$$\Delta U_{\text{sys}} = U_{\text{final}} - U_{\text{initial}}$$



Path functions

- Function whose value depends on the pathway (or sequence of steps) taken between the initial and final states.
- Examples: q and w

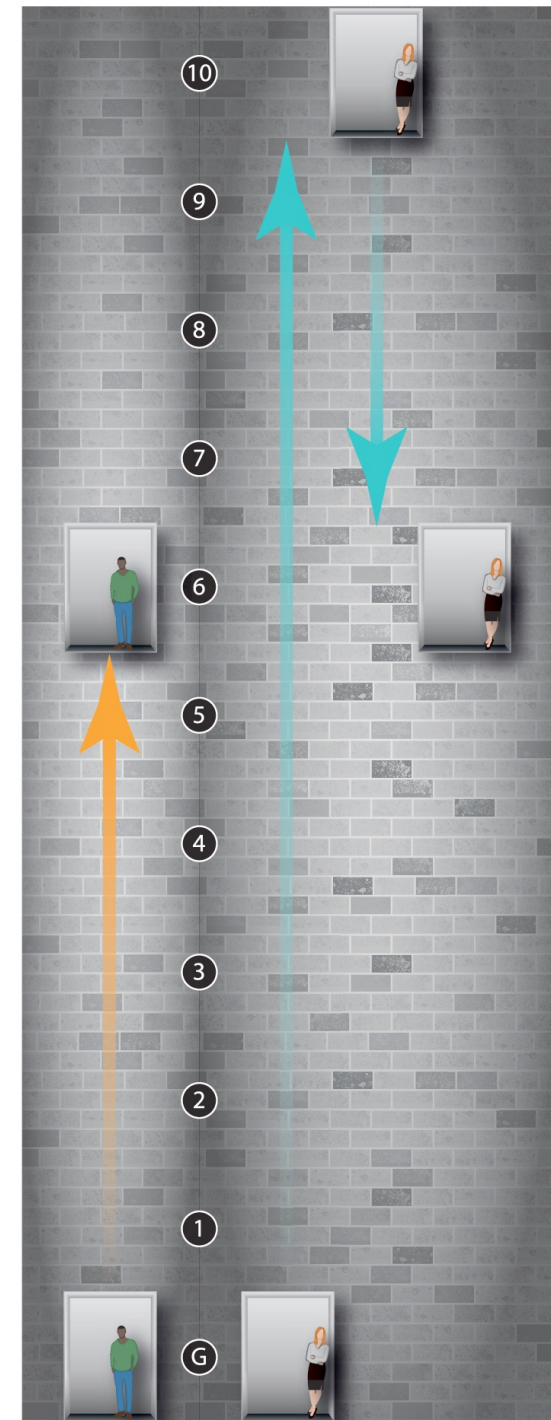


FIGURE 6.7 Vertical *displacement* is a state function, but total *distance traveled* is a path function.

Energy and Enthalpy

6.4

You may already be familiar with *mechanical work* associated with the *physical* movement of *macroscopic* objects. For example, lifting a book, pushing a crate, etc.

However, what is work (w) when it comes to *chemical* reactions/processes?

- In *most* chemical reactions, *only* pressure-volume (or PV) work is possible.
- PV work (w) is that associated with the expansion/compression (i.e., the change in volume, ΔV) against an external pressure due to the *net* production/consumption of *gaseous* species (Δn_{gas}) based on the balanced chemical equation.
- At a **constant pressure**: $w = -P \cdot \Delta V$ common units: P in atm, V in L, $1 \text{ L} \cdot \text{atm} = 101.325 \text{ J}$

Net Change in Number of Moles of Gas (Δn_{gas})	Change in Volume (ΔV)	Process	Energy of System	Sign of w
More gases in reactants $\Delta n_{\text{gas}} < 0$	Compression $\Delta V < 0$	Work <i>done on</i> the system	System <u>gains</u> energy	$w > 0$ $\oplus$
More gases in products $\Delta n_{\text{gas}} > 0$	Expansion $\Delta V > 0$	Work <i>done by</i> the system	System <u>loses</u> energy	$w < 0$ $\ominus$

IN-CLASS QUESTION 2

Learning Objective(s): 6.6

The ΔU_{sys} change in internal energy for a given system is -325 J . This system loses 155 J of energy as heat as it expands from an initial volume of 15.0 L at a constant pressure of 1.00 atm . $q < 0$
 $w < 0$ if expansion b/c $V_f > V_i$

What is the final volume (in L) of the system?

From the first law: $\Delta U_{\text{sys}} = q + w$; $q = -155 \text{ J}$

$$-325 \text{ J} = -155 \text{ J} + w$$

$$w = -170 \text{ J}$$

$$; w = -P \cdot \Delta V = -P \cdot (V_f - V_i)$$

$$-P \cdot (V_f - V_i) = -170 \text{ J}$$

$$-(1.00 \text{ atm}) \cdot (V_f - 15.0 \text{ L}) = -170 \text{ J} \times \frac{1 \text{ L} \cdot \text{atm}}{101.325 \text{ J}}$$

$$-(1.00 \text{ atm}) \cdot (V_f - 15.0 \text{ L}) = -1.677 \text{ L} \cdot \text{atm}$$

$$V_f = \frac{-1.677 \text{ L} \cdot \text{atm}}{-1.00 \text{ atm}} + 15.0 \text{ L}$$

$$V_f = 16.7 \text{ L}$$

TIP: fix/match units before algebra

work is energy, so $\text{L} \cdot \text{atm}$ is a unit of energy
 $1 \text{ L} \cdot \text{atm} = 101.325 \text{ J}$

Energy and Enthalpy

6.4

Most chemical experiments and reactions are best described as *open systems*.

- Exchange of *both* **heat** (q) and **work** (w) between the system and surroundings.
- Chemical reactions are often carried out on the “benchtop” where they are exposed to a **constant (atmospheric) pressure**, so

$$w = -P \cdot \Delta V \neq \text{zero}$$

- These conditions give rise to a very *useful and measurable* property called **enthalpy**.



Enthalpy (H) is a state function relating internal energy (ΔU), pressure (P), and volume (V).

From first law: $\Delta U = q + w$; if P_{constant} : $q \rightarrow q_p$, $w = -P \cdot \Delta V$

$$\Delta U = q_p - P \cdot \Delta V$$

$$q_p = \Delta U + P \cdot \Delta V \quad ; \quad \text{define right side as } \Delta H \equiv \Delta U + P \cdot \Delta V$$

$$q_p = \Delta H$$

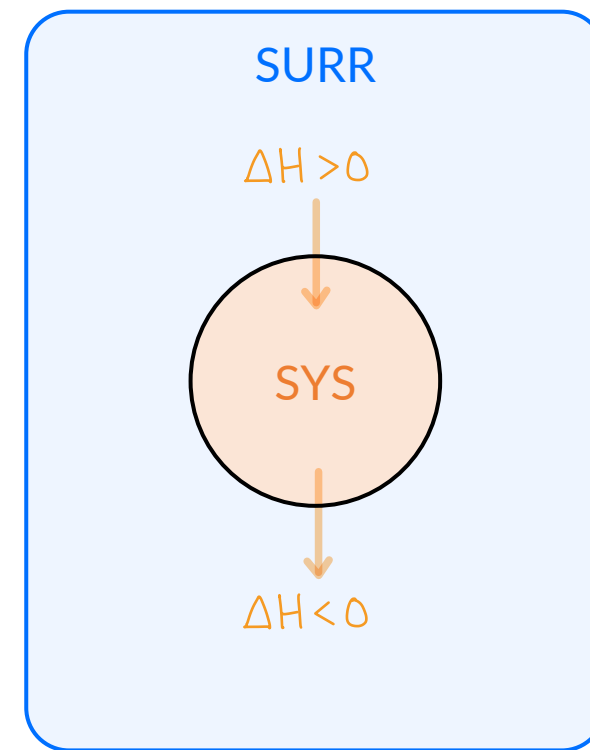
Energy and Enthalpy

6.4

The change in enthalpy for a reaction (ΔH_{rxn}) is *defined* as the flow of heat (q) when the chemical reaction is carried at a constant (atmospheric) pressure.

$q_{p,\text{rxn}} \leftrightarrow \Delta H_{\text{rxn}}$ synonymous at constant pressure!

Process	System/Rxn	Surroundings	Sign of ΔH_{rxn}
Exothermic	Releases heat $T_{\text{sys}} \downarrow$	Absorbs heat $T_{\text{sur}} \uparrow$	$\Delta H_{\text{rxn}} < 0$ $\ominus$
Endothermic	Absorbs heat $T_{\text{sys}} \uparrow$	Releases heat $T_{\text{sur}} \downarrow$	$\Delta H_{\text{rxn}} > 0$ $\oplus$



BONUS QUESTION

Assessed on: Fall 2021, Exam 2

What is the change in internal energy (in Joules) if a system gains 0.886 kJ of heat upon expanding from 2.123 L to 10.688 L against a constant pressure of 2.86 atm?